Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A particle of charge q and mass m moves in a circular orbit of radius r with the angular speed ω. The ratio of the magnitude of its magnetic moment to that of its angular momentum depends on

(A) ω and q

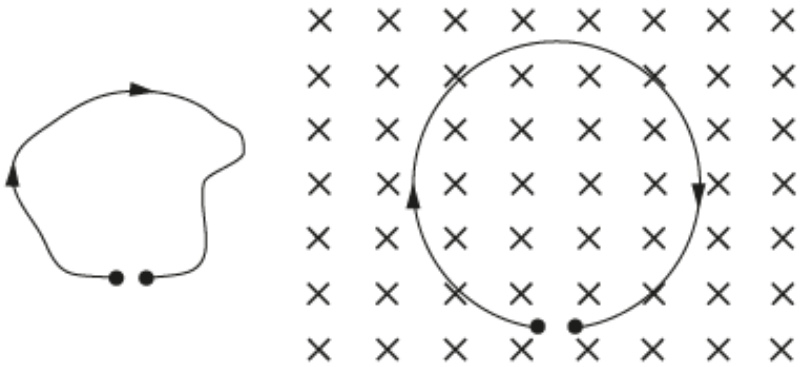
(B) ω, q and m

(C) q and m

(D) ω and m

Q.2

A thin flexible wire of length L is connected to two adjacent fixed points and carries a current I in the clockwise direction, as shown in the figure. When the system is put in a uniform magnetic field of strength B going into the plane of the paper, the wire takes the shape of a circle. The tension in the wire is



(A) IBL

(B) $\frac{IBL}{\pi}$

(C) $\frac{IBL}{2\pi}$

(D) $\frac{IBL}{4\pi}$

Q.3

A long straight wire along the z-axis carries a current I in the negative z-direction. The magnetic vector field $\vec{B}$ at a point having the coordinates (x, y) in the $z = 0$ plane is

(A) $\left(\frac{\mu_0 I}{2\pi}\right)\left(\frac{y\vec{i}-x\vec{j}}{x^2+y^2}\right)$

(B) $\left(\frac{\mu_0 I}{2\pi}\right)\left(\frac{x\vec{i}+y\vec{j}}{x^2+y^2}\right)$

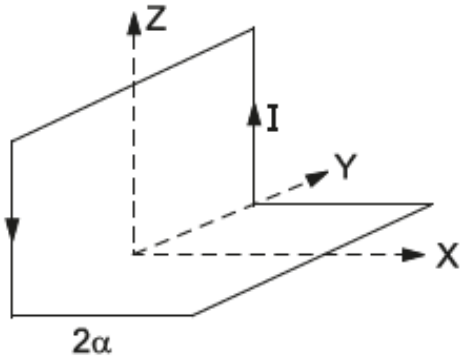
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(C) $\left(\frac{\mu_0 I}{2\pi}\right)\left(\frac{x\vec{j}-y\vec{i}}{x^2+y^2}\right)$

(D) $\left(\frac{\mu_0 I}{2\pi}\right)\left(\frac{x\vec{i}-y\vec{j}}{x^2+y^2}\right)$

Q.4

A nonplanar loop of a conducting wire carrying a current I is placed as shown in the figure. Each of the straight sections of the loop is of length $2a$. The magnetic field due to this loop at the point $P(a, 0, a)$ points in the direction



(A) $\frac{1}{\sqrt{2}}(-\vec{j}+\vec{k})$

(B) $\frac{1}{\sqrt{3}}(-\vec{j}+\vec{k}+\vec{i})$

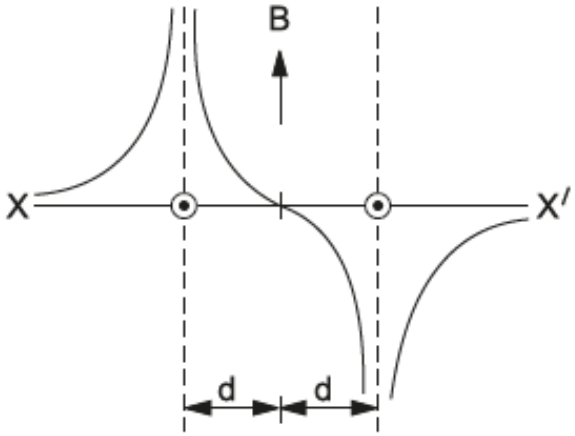
(C) $\frac{1}{\sqrt{3}}(\vec{i}+\vec{j}+\vec{k})$

(D) $\frac{1}{\sqrt{2}}(\vec{i}+\vec{k})$

Q.5

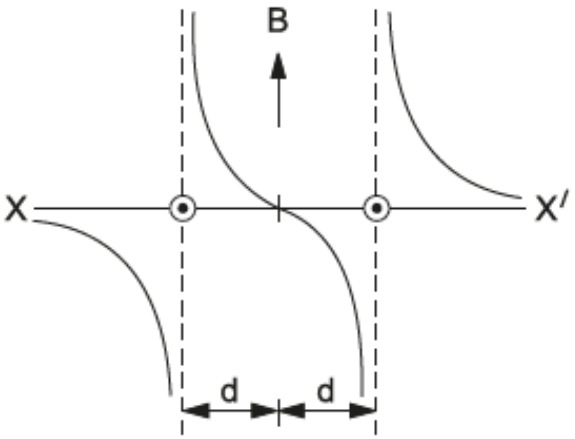
Two long parallel wires are at a distance $2d$ apart. They carry steady equal currents flowing out of the plane of the paper, as shown. The variation of the magnetic field B along the line XX' is given by

(A)

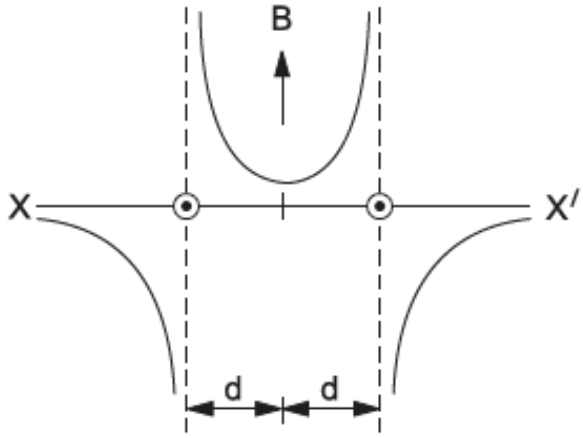


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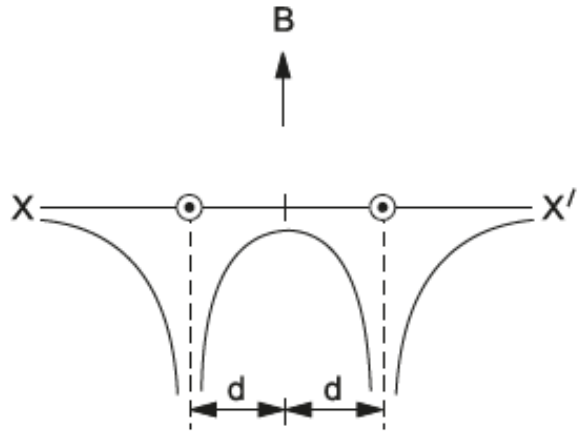
(B)



(C)

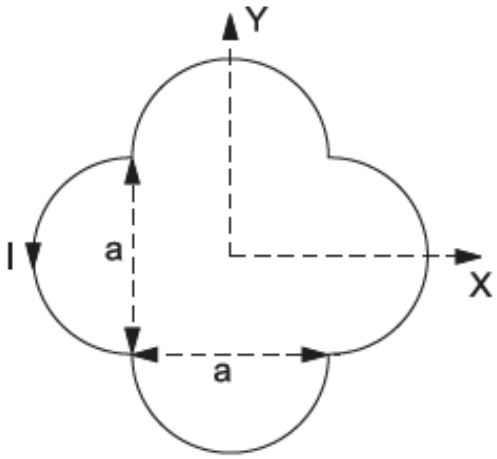


(D)



Q.6

A loop carrying a current I lies in the xy-plane as shown in the figure. The unit vector $\hat{k}$ is coming out of the plane of the paper. The magnetic moment of the current loop is



(A) $a^2 I\hat{k}$

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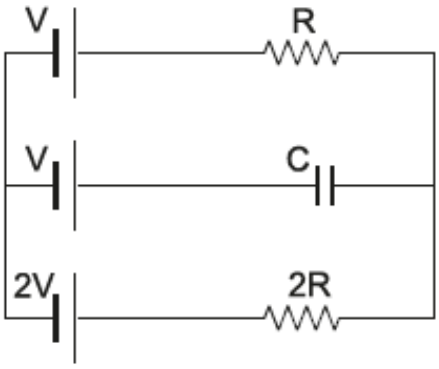
(B) $\left(\frac{\pi}{2}+1\right) a^2 I\hat{k}$

(C) $-\left(\frac{\pi}{2}+1\right) a^2 \hat{I}\hat{k}$

(D) $(2\pi+1)a^2 I\hat{k}$

Q.7

In the given circuit with a steady current, the potential drop across the capacitor must be



(A) V

(B) $\frac{V}{2}$

(C) $\frac{V}{3}$

(D) $\frac{2V}{3}$

Q.8

A particle of mass m and charge q moves with a constant velocity v along the positive x-direction. It enters a region containing a uniform magnetic field B directed along the negative z-direction, extending from $x = a$ to $x = b$. The minimum value of v required so that the particle can just enter the region $x > b$ is

(A) $\frac{qbB}{m}$

(B) $\frac{q(b-a)B}{m}$

(C) $\frac{qaB}{m}$

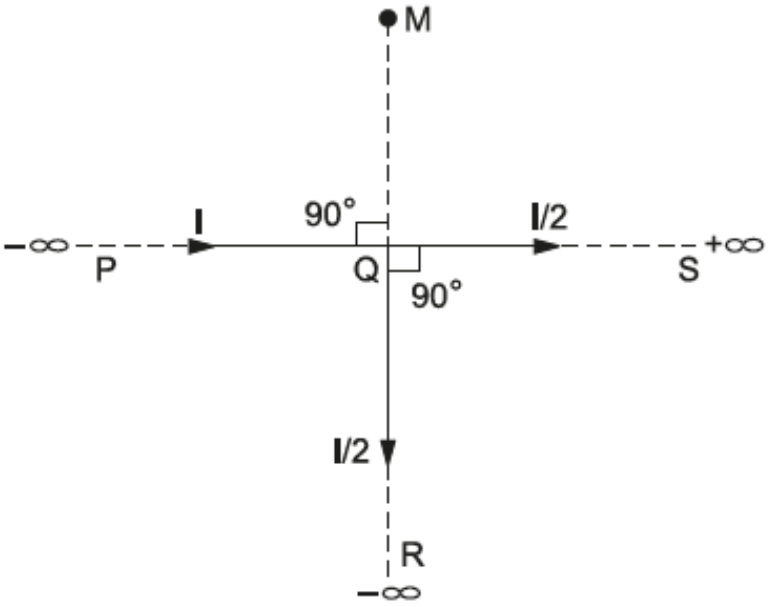
(D) $\frac{q(b+a)B}{2m}$

Q.9

An infinitely long conductor PQR is bent to form a right angle as shown. A current I flows through PQR. The magnetic field strength due to this current at the point M is H_1. Now, another infinitely long straight conductor QS is connected at Q so that the current is $I/2$ in QR as well as in QS, the current in PQ

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remaining unchanged. The magnetic field strength at M is now H_2. The ratio $\mathrm{H}_1/\mathrm{H}_2$ equals



(A) $\frac{1}{2}$

(B) 1

(C) $\frac{2}{3}$

(D) 2

Q.10

A coil of 50 turns and $10\ \mathrm{cm}$ diameter has a resistance of 10Ω. What must be the potential difference across the coil so as to nullify the earth's magnetic field $H = 0.314\mathrm{G}$ at the centre of the coil?

(A) $0.5\ \mathrm{V}$

(B) $1.0\ \mathrm{V}$

(C) $1.5\ \mathrm{V}$

(D) $2.0\ \mathrm{V}$

Q.11

The magnetic field at the point of intersection of diagonals of a square loop of side L carrying a current I is

(A) $\frac{\mu_0 I}{\pi L}$

(B) $\frac{2\mu_0 I}{\pi L}$

(C) $\frac{\sqrt{2}\mu_0 I}{\pi L}$

(D) $\frac{2\sqrt{2}\mu_0 I}{\pi L}$

Q.12

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Two identical coils carry equal currents and have a common centre, but their planes are at right angles to each other. What is the magnitude of the resultant magnetic field at the centre if the field due to one coil alone is B ?

(A) zero

(B) $B/\sqrt{2}$

(C) $\sqrt{2}B$

(D) $2B$

Q.13

The direction of the force experienced by a charged particle moving with a velocity $\mathbf{v}$ in a uniform magnetic field $\mathbf{B}$ is

(A) parallel to $\mathbf{v}$ and perpendicular to $\mathbf{B}$

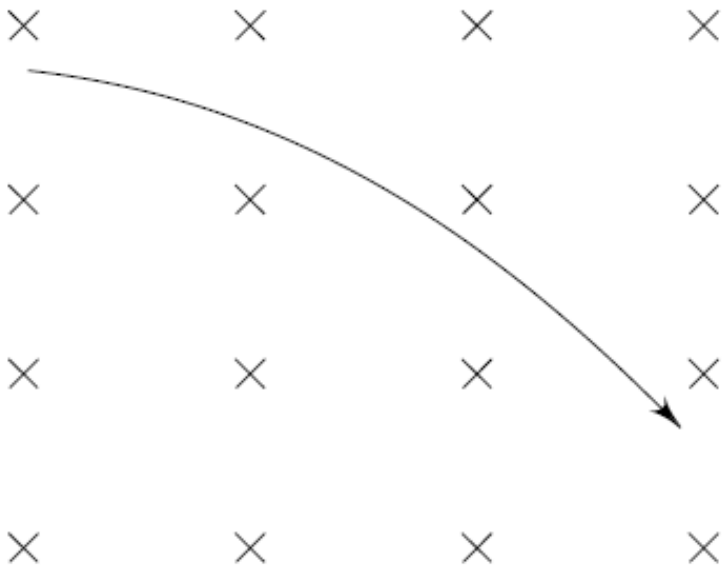
(B) parallel to $\mathbf{B}$ and perpendicular to $\mathbf{v}$

(C) parallel to both $\mathbf{v}$ and $\mathbf{B}$

(D) perpendicular to both $\mathbf{v}$ and $\mathbf{B}$.

Q.14

A particle is projected into a uniform magnetic field acting perpendicular to the plane of the paper. The field points into the paper, indicated by $\times$ which represents the tail of the field vector. The trajectory shown could be that of a (see Fig.)



(A) neutron

(B) proton

(C) alpha particle

(D) electron

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Q.15

A proton with kinetic energy K describes a circle of radius r in a uniform magnetic field. An α-particle with kinetic energy K moving in the same magnetic field will describe a circle of radius

(A) $\frac{r}{2}$

(B) r

(C) $2r$

(D) $4r$

Q.16

An α-particle moving with a velocity v in a uniform magnetic field is moving in a circular path at frequency v called the cyclotron frequency. The cyclotron frequency of a proton moving with a speed $2v$ in the same magnetic field will be

(A) $\frac{v}{4}$

(B) $\frac{v}{2}$

(C) v

(D) $2v$

Q.17

A conducting circular loop of radius r carries a constant current i. It is placed in a uniform magnetic field B such that B is perpendicular to the plane of the loop. The magnetic force acting on the loop is

(A) ir B

(B) $2\pi irB$

(C) zero

(D) π i rB

Q.18

For a paramagnetic material, the dependence of the magnetic susceptibility χ on the absolute temperature T is given by (C is a constant)

(A) $\chi = CT$

(B) $\chi = C/T$

(C) $\chi = CT^2$

(D) $\chi = CT^{-2}$

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Q.19

A small piece of a substance is repelled when it is brought near a powerful magnet. What can you say about its type ?

(A) dimagnetic

(B) paramagnetic

(C) ferromagnetic

(D) non-magnetic

Q.20

A 2MeV proton is moving perpendicular to a uniform magnetic field of $2.5\ \text{T}$. The force on the proton is

(A) $2.5 \times 10^{-10}\ \text{N}$

(B) $8 \times 10^{-11}\ \text{N}$

(C) $2.5 \times 10^{-11}\ \text{N}$

(D) $8 \times 10^{-12}\ \text{N}$

Q.21

The relative permeability of iron is of the order of 10^x. The value of x is

Q.22

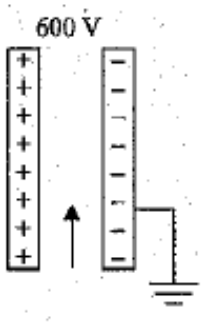
A sample of paramagnetic salt contains 2×10^{24} atomic dipoles, each of dipole moment 1.5×10^{-23} JT^{-1}. The sample is placed in a magnetic field of $0.6\ \text{T}$ and cooled to a temperature of $4\ \text{K}$. The degree of magnetic saturation achieved is equal to 15%. What is the total dipole moment of the sample for a magnetic field of $0.9\ \text{T}$ and a temperature of $3\ \text{K}$? (in J/T)

Q.23

A potential difference of $600\ \text{V}$ is applied across the plates of a parallel plate capacitor placed in a magnetic field. The separation between the plates is $\mathbf{3\ mm}$. An electron projected vertically upward, parallel to the plates, with a velocity of $2\times\ 10^6\ \text{ms}^{-1}$ moves undeflected between the plates. Find the magnitude and direction of the magnetic field in the region between the capacitor plates. Find the magnitude and direction of the magnetic field in the region between the capacitor plates. Given the Charge of the electron $= -1.6 \times$

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10^{-19}C.



Q.24

A proton of mass m and charge $+e$ is moving in a circular orbit a magnetic field with energy 1MeV. What should be the energy of an α-particle (mass 4 m and charge $+2e$) so that it revolves in a circular orbit of the same radius in the same magnetic field? (in MeV)

Q.25

At a certain place, the horizontal component of earth's magnetic field is $\sqrt{3}$ times the vertical component. The angle of dip at that place is (in degrees)

Q.26

The magnetic field due to a current carrying circular loop of radius 3 cm at a point on its axis at a distance of 4 cm from the centre is $54\mu\text{T}$. The magnetic field (in μT) at the centre of the loop will be

Q.27

If in a circular coil A of radius R, current i is flowing and in another coil B of radius $2R$, current $2i$ is flowing, then the ratio of magnetic fields B_A and B_B at their centre is

Q.28

A particle of charge $-1.6 \times 10^{-19}\text{C}$ moving with velocity 10 m s^{-1} along the x-axis enters a region where the magnetic field $\mathbf{B}$ is along the y-axis and an electric field of magnitude 10^4 V m^{-1} is along the negative z-axis. If the charged particle continuous moving along the x-axis, the magnitude of $\mathbf{B}$ is

Q.29

A long straight wire of radius a carries a steady current i. The current is uniformly distributed across its cross-section. The ratio of the magnetic field at $\frac{a}{2}$ and $2a$ is

Q.30

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A charge q moves with a velocity v in a magnetic field $\mathbf{B} = \hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 4\hat{\mathbf{k}}$. If the acceleration of the particle at time t is $\mathbf{a} = 2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} - n\hat{\mathbf{k}}$, then the value of n is

Answers & Solutions

Q.1 Answer:

q and m

Solution:

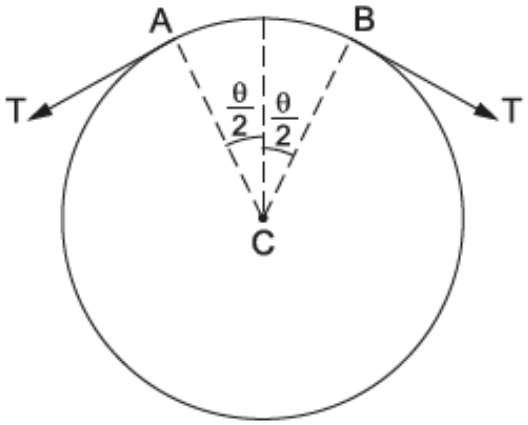
The effective current is $i = q \cdot \frac{\omega}{2\pi}$ and the area is $A = \pi r^2$
The magnetic moment is $\mu = Ai = \frac{1}{2}q\omega r^2$
The angular momentum is $L = I\omega = mr^2\omega$. $\therefore \frac{\mu}{L} = \frac{q}{2m}$

Q.2 Answer:

$\frac{IBL}{2\pi}$

Solution:

$L = 2\pi r \quad \Rightarrow \quad r = \frac{L}{2\pi}$
(Length of the element AB) $= dl = r\theta = \left(\frac{L}{2\pi}\right)\theta$
(Outward ampere force on AB) $\Rightarrow$ BI dl.
(Inward force on AB due to tension T)
$= 2T\sin(\theta/2) = T\theta$
Equating each of the two forces for equilibrium $T = \frac{BIL}{2\pi}$.



Q.3 Answer:

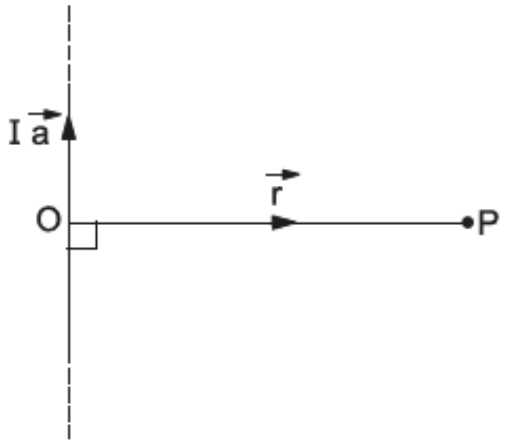
$\left(\frac{\mu_0 I}{2\pi}\right)\left(\frac{y\vec{i} - x\vec{j}}{x^2+y^2}\right)$

Solution:

To find the magnetic field at a point P due to a long straight wire carrying the current I, let $\vec{a}$ be a unit vector in the direction of current flow, and $\vec{r}$ be the vector, normal to $\vec{a}$, from the wire to P. Then the magnetic field at P is

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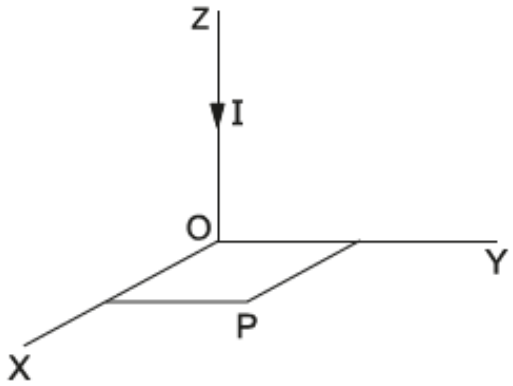
$\vec{B} = \frac{\mu_0 I}{2\pi} \cdot \frac{\vec{a} \times \vec{r}}{r^2}$



Here, $\vec{a} = -\vec{k}$ **and** $\vec{r} = x\vec{i} + y\vec{j}$

$$\therefore \quad \vec{B} = \frac{\mu_0 I}{2\pi} \cdot \frac{(-\vec{k}) \times (x\vec{i} + y\vec{j})}{r^2}$$

$$= \left(\frac{\mu_0 I}{2\pi}\right)\left(\frac{y\vec{i} - x\vec{j}}{x^2 + y^2}\right)$$



Q.4 Answer:

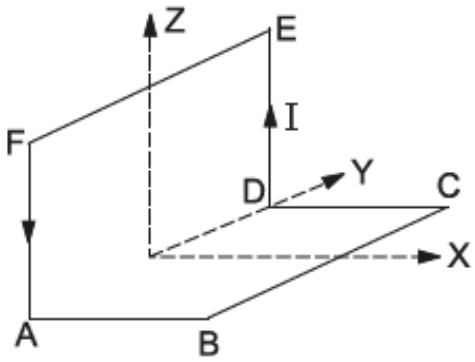
$\frac{1}{\sqrt{2}}(\vec{i} + \vec{k})$

Solution:

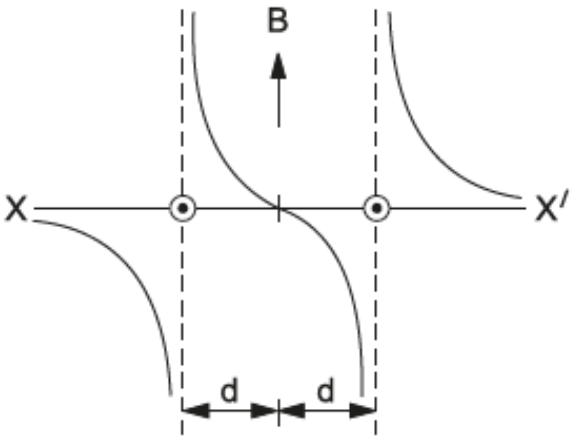
The conductors BC **and** EF **will together produce a magnetic field at** P**, which lies in the** xz**-plane and is equally inclined to the** x **- and** z**-axes. The conductors** AB **and CD will together produce a magnetic field at** P **directed along the** z**-axis and of the magnitude** B_0 **(say). The conductors DE and FA will produce a field at** P **with the same magnitude** B_0**, and directed along the** x**-axis. Thus, the field at P will have equal components in the** x**-and** z**-directions, and no component in the** y**-direction.**

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It must therefore point in the direction $(1/\sqrt{2})(\vec{i}+\vec{k})$



Q.5 Answer:



Solution:

The magnetic field lines due to a straight current constitute concentric circles. Hence, the magnetic field between the two wires act in opposite directions. Similarly, the magnetic field to the left of the first wire and that to the right of the second wire are oppositely directed. Hence the correct graph is (b).

Q.6 Answer:

$\left(\frac{\pi}{2}+1\right) a^2 I\hat{k}$

Solution:

Area of the current loop, $A = a^2 + 4 \times \frac{\pi(a/2)^2}{2} = a^2 + \frac{\pi a^2}{2}$.

$\therefore$ magnetic moment $= \vec{m} = \overrightarrow{IA} = I\left(\frac{\pi}{2}+1\right) a^2\hat{k}$

Q.7 Answer:

$\frac{V}{3}$

Solution:

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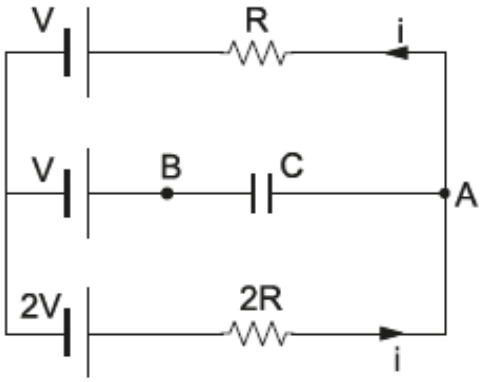
Moving anticlockwise from A,

$$iR + V - 2V + 2iR = 0$$

$$\text{or} \quad 3iR = V \quad \text{or} \quad i = \frac{V}{3R}$$

$$\text{Then, } V_A - V_B = iR + V - V = iR$$

$$= \frac{V}{3} = \text{ potential drop across } C$$



Q.8 Answer:

$\frac{q(b-a)B}{m}$

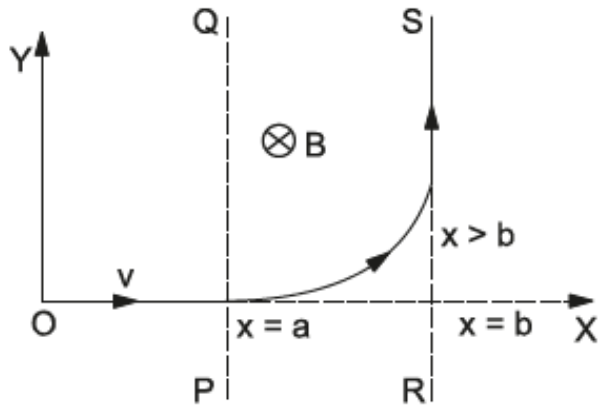
Solution:

In the figure, the z-axis points out of the paper, and the magnetic field is directed into the paper, existing in the region between PQ and RS. The particle moves in a circular path of radius r in the magnetic field. It can just enter the region $x > b$ for $r \geq b - a$

or $r = \frac{mv}{Bq} \geq b - a$

or $v \geq \frac{q(b-a)B}{m}$

or $v_{\min} = \frac{q(b-a)B}{m}$



Q.9 Answer:

$\frac{2}{3}$

Solution:

The magnetic field strength at M due to QR is zero in all cases. When QS carries $I/2$ current, the field strength at M due to QS is $H_1/2$. Then

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$H_2 = H_1 + \frac{H_1}{2} = \frac{3H_1}{2}$ or $\frac{H_1}{H_2} = \frac{2}{3}$

Q.10 Answer:

0.5 V

Solution:

$H = 0.314G = 0.314 \times 10^{-4}$ T. **Let the current in coil be** I**. Then the magnetic field at the centre of the coil is**

$$B = \frac{\mu_0 In}{2r} = \frac{4\pi \times 10^{-7} \times I \times 50}{2 \times 5 \times 10^{-2}}$$
$$= 2\pi I \times 10^{-4} \text{ T}$$

The value of I **for which** $B = H$ **is given by** $2\pi I \times 10^{-4} = 0.314 \times 10^{-4}$

or $I = 0.05$ A

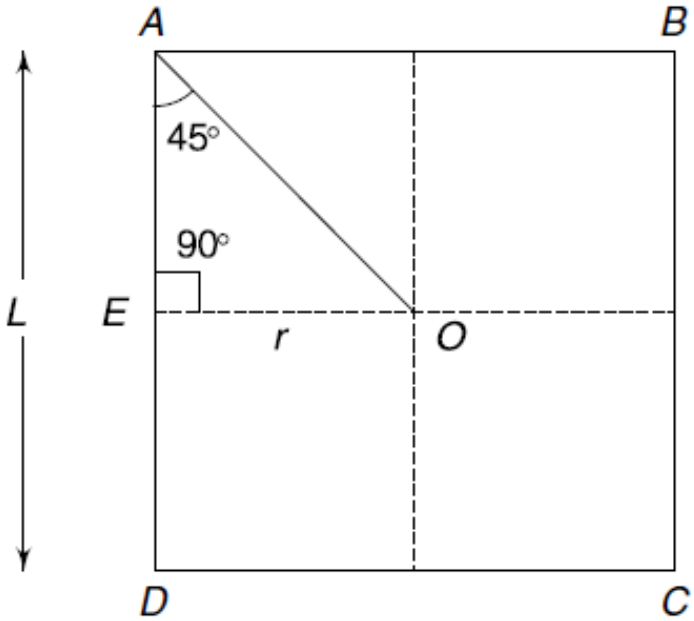
$\therefore$ **Potential difference** $= IR = 0.05 \times 10 = 0.5$ V **Hence the correct choice is (a).**

Q.11 Answer:

$\frac{2\sqrt{2}\mu_0 I}{\pi L}$

Solution:

Refer to Fig. Here $r = OE = \frac{L}{2}$



The magnetic field at the centre O **due to the current element** AE **is given by**

$B_{AE} = -\frac{\mu_0 I}{4\pi r} \int_{90^\circ}^{45^\circ} \sin\theta d\theta$

$= \frac{\mu_0 I}{4\pi r} |\cos\theta|_{90^\circ}^{45^\circ}$

$= \frac{\mu_0 I}{4\pi r} (\cos 45^\circ - \cos 90^\circ)$

$= \frac{\mu_0 I}{4\pi r} (\cos 45^\circ - 0) = \frac{\mu_0 I}{4\sqrt{2}\pi r}$

It is clear that the magnetic field at O **due to current element** DE **is the same as that due to** AE**.**

Hence, the magnetic field at O **due to one side** AD **is**

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$B_{AD} = \frac{2\mu_0 I}{4\sqrt{2}\pi r} = \frac{\sqrt{2}\mu_0 I}{4\pi r}$

Since the centre of the square is equidistant from the ends A, B, C and D of each side of the square and each side produces at the centre O the same magnetic field, the field due to the square is 4 times that due to one side. Hence (because $r = L/2$)

$B = 4B_{\mathrm{AD}} = \frac{\sqrt{2}\mu_0 I}{\pi r} = \frac{2\sqrt{2}\mu_0 I}{\pi L}$

Hence the correct choice is (d).

Q.12 Answer:

$\sqrt{2}B$

Solution:

The magnitude of the magnetic field at the centre due to each coil is B. Since the planes of the coils are at right angles to each other, the directions of the fields will be at right angles to each other. Therefore, the resultant field is

$B_r = \sqrt{B^2 + B^2} = \sqrt{2}B$

Hence the correct choice is (c).

Q.13 Answer:

perpendicular to both $\mathbf{v}$ and $\mathbf{B}$.

Solution:

The correct choice is (d) since $\mathbf{F} = q(\mathbf{v} \times \mathbf{B})$

Q.14 Answer:

electron

Solution:

Applying the left hand fist rule and noting the charge on an electron is negative, i.e. $\mathbf{F} = -e(\mathbf{v} \times \mathbf{B})$, it follows that the particle is an electron, which is choice (d).

Q.15 Answer:

r

Solution:

The radius of the circular orbit is given by

$r = \frac{\sqrt{2mK}}{qB}$

The charge of an α-particle is twice that of a proton and its mass is four times the mass of a proton. Therefore $\sqrt{m}/q$ is the same for both. Hence r will the same for both particles. Thus the correct choice is (b).

Q.16 Answer:

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$2v$

Solution:

The cyclotron frequency is given by

$v = \frac{qB}{2\pi m}$

It is independent of the speed of the particle and the radius of its circular path. Now $v \propto q/m$. The charge of a proton is half that of an α-particle and the mass of a proton is one-fourth. Therefore, v will be doubled. Hence the correct choice is (d).

Q.17 Answer:

zero

Solution:

The correct choice is (c) because the magnetic field produced by the current in the loop and the external magnetic field are along the same direction.

Q.18 Answer:

$\chi = C/T$

Solution:

According to Curie's law, the susceptibility χ is related to temperature T as

$\chi = \frac{C}{T}$

where C is curie constant. Hence the correct choice is (b).

Q.19 Answer:

dimagnetic

Solution:

Diamagnetic materials, like water, or water based materials, have a relative magnetic permeability that is less than or equal to 1 . Thus they have a magnetic susceptibility. Hence they can be repelled by magnetic fields.

Q.20 Answer:

8×10^{-12} N

Solution:

The kinetic energy of proton is

$$K = 2\text{MeV} = 2 \times 10^6 \text{eV}$$
$$= 2 \times 10^6 \times 1.6 \times 10^{-19} \text{ J} = 3.2 \times 10^{-13} \text{ J}$$

$$\therefore \quad \frac{1}{2}mv^2 = 3.2 \times 10^{-13}$$

Now, mass of proton is $m = 1.67 \times 10^{-27}$ kg. Therefore,

$v^2 = \frac{2 \times 3.2 \times 10^{-13}}{1.67 \times 10^{-27}} = 3.83 \times 10^{14}$

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or $v = 1.96 \times 10^7\ \text{ms}^{-1}$. Now force on proton is

$$F = evB$$
$$= 1.6 \times 10^{-19} \times 1.96 \times 10^7 \times 2.5$$
$$= 7.84 \times 10^{-12}\ \text{N}$$

Hence the closest choice is (d).

Q.21 Answer:

—

Solution:

Relative permeability of iron is 5500. It is order of 3.

Q.22 Answer:

—

Solution:

Initial total magnetic moment at temperature $T = 4\ \text{K}$ is

$m_1 = 15\%$ of $\left(2.0 \times 10^{24} \times 1.5 \times 10^{-23}\right) = 4.5\text{JT}^{-1}$

Now from Curie's law, we have $\chi = C/T$ or $\frac{I}{H} = \frac{C}{T}$

where I is magnetization and H is the magnetizing field. If V is the volume of the sample, then by definition, $M = I/V$ and Eq. (i) becomes

$M = \frac{C}{V}\left(\frac{H}{T}\right) = \text{ constant } \times \left(\frac{H}{T}\right)$

If M_2 is the magnetic moment at temperature $T_2 = 3\ \text{K}$ and field $H_2 = 0.9\ \text{T}$, then from Eq. (ii) M_1 and M_2 are related as

$\frac{M_1}{M_2} = \frac{H_1}{H_2} \cdot \frac{T_2}{T_1}$

$$M_2 = M_1 \times \frac{H_2}{H_1} \times \frac{T_1}{T_2}$$
$$\text{or } = 4.5 \times \frac{0.9}{0.6} \times \frac{4}{3} = 9\text{JT}^{-1}$$

Q.23 Answer:

—

Solution:

The force on electron will be toward the left plane due to electric field and will be equal to $\text{F}_\text{e} = \text{eE}$. For the electron to move undeflected between the plates, there should be a force (magnetic) which is equal to the electric force and opposite in direction to electric force. So the force should be directed toward the right as the electric force is toward the left. On applying Fleming's left hand rule, we get that the magnetic field should be directed perpendicular to the plane of paper and inwards.

Therefore,

Force due to electric field = Force due to magnetic field

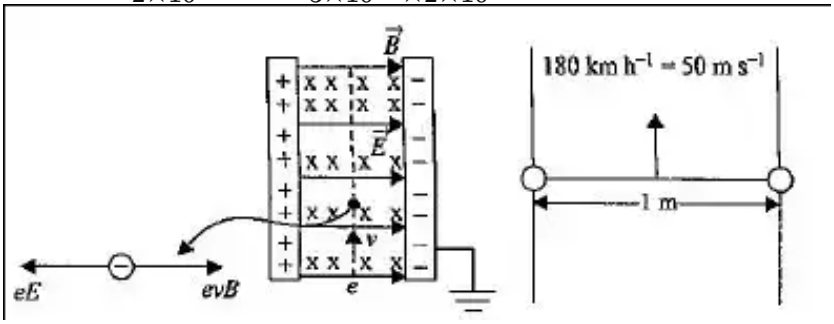
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$eE = evB \left[\because E = \frac{V}{d}\right]$

$B = \frac{E}{v} = \frac{V/d}{v}$

where, V = potential difference between the plates, and d = distance between the plates

$B = \frac{600/3\times10^{-3}}{2\times10^{6}} = \frac{600}{3\times10^{-3}\times2\times10^{6}} \Rightarrow B = 0.1\ \text{T}$



Q.24 Answer:

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Solution:

For proton: $r = \frac{mv}{eB}$

For α-particle $r' = \frac{m'v'}{e'B} = \frac{4mv'}{2eB} = \frac{2mv'}{eB}$

Given $r = r'$. **Hence** $v' = \frac{v}{2}$.

Energy of proton $E = \frac{1}{2}mv^2$. **Energy of α-particle is**

$E' = \frac{1}{2}m'v'^2 = \frac{1}{2} \times 4m \times \left(\frac{v}{2}\right)^2 = \frac{1}{2}mv^2 = E$

Hence $E' = 1\text{MeV}$

Q.25 Answer:

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Solution:

$B_H = B\cos\theta$ **and** $B_V = B\sin\theta$. **Hence**

$\frac{B_V}{B_H} = \tan\theta$

Given $\frac{B_V}{B_H} = \frac{1}{\sqrt{3}}$. **Therefore,** $\tan\theta = \frac{1}{\sqrt{3}}$, **i.e.** $\theta = 30^\circ$

Q.26 Answer:

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Solution:

Given $x = 4\ \text{cm} = 0.04\ \text{m}$ **and** $r = 3\ \text{cm} = 0.03\ \text{m}$. **The magnetic field at a point on the axis of the loop is given by**

$B = \frac{\mu_o I r^2}{2(r^2+x^2)^{3/2}}$ **(1)**

Magnetic field at the centre of the coil is given by

$B_0 = \frac{\mu_o I}{2r}$ **(2)**

Dividing (1) by (2), we get

$\frac{B_0}{B} = \frac{r^3}{(r^2+x^2)^{3/2}}$

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Substituting the values of r and x, we get

$\frac{B_0}{B} = \frac{125}{27}$

or $B_0 = \frac{125}{27}B = \frac{125}{27} \times 54\mu\text{T} = 250\mu\text{T}$

Q.27 Answer:

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Solution:

$B_A = \frac{\mu_0 i}{2R}$ and $B_B = \frac{\mu_0(2i)}{2(2R)}$

$\therefore \quad \frac{B_A}{B_B} = 1$

Q.28 Answer:

—

Solution:

$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$

Given $\mathbf{E} = -E\hat{\mathbf{k}}, \mathbf{v} = v\hat{\mathbf{i}}$ and $\mathbf{B} = B\hat{\mathbf{j}}$

$\therefore \mathbf{F} = q[-E\mathbf{k} + (v)\hat{\mathbf{i}}) \times (B\hat{\mathbf{j}})]$

$= q(-E\hat{\mathbf{k}} + vB\hat{\mathbf{k}}) = q(-E + vB)\hat{\mathbf{k}}$

Since the particle moves undeflected, $\mathbf{F} = 0$**, i.e.**

$q(-E + vB)\hat{k} = 0$

$\Rightarrow -E + vB = 0$

$\Rightarrow -EB = \frac{E}{v} = \frac{10^4}{10} = 10^3 \text{ T}$

Q.29 Answer:

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Solution:

The magnetic field at a distance r from the axis of the wire of radius a and carrying a current i for the case when r lies between 0 and a is given by

$$B_1 = \frac{\mu_0 i r}{2\pi a}; \qquad 0 \le r \le a$$

$$= \frac{\mu_0 i}{2\pi a} \times \frac{a}{2} \quad \left(\text{for } r = \frac{a}{2}\right)$$

$$\Rightarrow \quad B_1 = \frac{\mu_0 i}{4\pi a}$$

For the case $r > a$, the magnetic field is given by

$$B_2 = \frac{\mu_0 i}{2\pi r} = \frac{\mu_0 i}{2\pi(2a)} \quad (\text{ for } r = 2a)$$

$$\therefore \quad \frac{B_1}{B_2} = 1$$

Q.30 Answer:

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Solution:

$\mathbf{F} = q(\boldsymbol{v} \times \mathbf{B})$. **Hence** $\mathbf{F}$ **is perpendicular to** $\mathbf{B}$ **Therefore,**

$$\begin{aligned}
\mathbf{F}.\mathbf{B} &= 0 \\
\text{or } m\mathbf{a} \cdot \mathbf{B} &= 0 \\
\Rightarrow \mathbf{a} \cdot \mathbf{B} &= 0 \quad (\because m \neq 0) \\
\Rightarrow (2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} - n\hat{\mathbf{k}})^{\circ}(\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 4\hat{\mathbf{k}}) &= 0 \\
\Rightarrow 2 + 6 - 4n &= 0 \\
\Rightarrow n &= 2
\end{aligned}$$

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